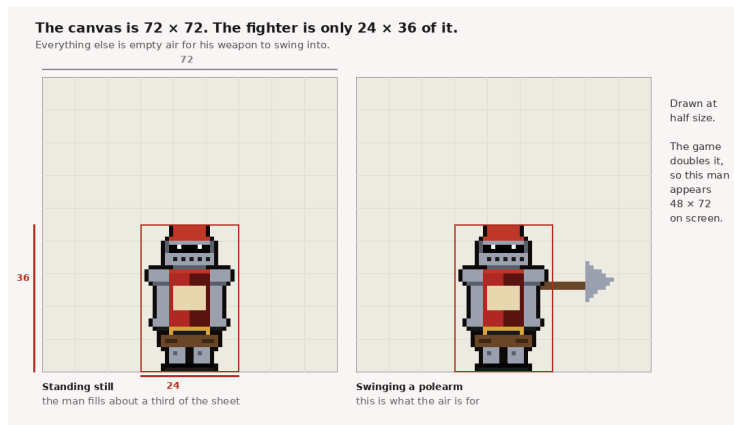


# Retro Buhurt Art Specification

BonkWorks · 14 September 2026 · the brief, and the Retro Bowl teardown it was derived from

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## Start here — the whole job in plain English



**The canvas is 72 × 72. The man is only 24 × 36 of it.** The rest is empty air for his weapon to swing through. **If you fill the 72 with a man, every fighter comes out twice the size the game expects.**

### The nine things to get right

1. **Draw everything at half the size it appears.** The game doubles it on screen, so a 24 × 36 man shows up 48 × 72. That doubling is the retro look.
2. **The size is 8-bit, the colour isn't.** A fighter is about a thumbnail tall on a phone — the NES part. But he gets **twelve colours a frame** where a real NES allowed four.
3. **Hard pixels only.** No soft edges, no blur, no gradients. **Turn anti-aliasing off** — the game recolours these sprites at runtime and that only works on solid pixels. A soft edge renders in-game as bright magenta.
4. **Nine of the twelve colours are placeholders.** Surcoat, helm, steel and leather are painted in flat nonsense colours — pure magenta, pure red, pure green — and the game swaps them for each club's real colours as it draws. One drawing serves every team in the league. **Type the hex values in exactly.** PNG only.
5. **The other three are fixed:** black, near-black, and one white highlight. One.
6. **A one-pixel black outline around the whole man.** Two armoured men standing close together fuse into one grey blob without it.
7. **Nobody stabs.** Every attack is a swing. Thrusting is illegal in buhurt.
8. **Nobody gets back up.** Once a man is on the ground he is out. There is no standing-up animation.
9. **Don't draw a blow landing on** a neck, groin, spine, back of the head, back of the knee or a foot. All illegal targets.

The job, in one line

**16 files, 52 frames.** Two fighters — one with a falchion or axe and a shield, one with a polearm — each doing eight things: standing guard, walking, charging, swinging, grappling, throwing, catching his breath, and lying down. **Facing right only;** the game mirrors him for the other direction. Frames sit side by side in one strip with no gaps between them.

Everything after this page is the evidence and the exact numbers. The full asset table, the colour keys and the canvas rules are in §8. If you only read two things, read this page and that section.

Every number in this document was parsed out of Retro Bowl's own asset manifest — `RetroBowl.js` (3.0 MB) and its two texture atlases — not read off a community guide. That gives exact canvas sizes, frame counts, playback speeds and draw origins for all **205 sprites and 632 frames**, the five fonts by name, and the full source of the shader that puts a team's colours on a player.

Where this document says a thing *is* a size, that size came out of the manifest. Where it reads intent into a decision, it says so. The last section maps all of it onto Retro Buhurt as the code stands today, and lists the four places the two designs currently disagree.

How to read this

The sections alternate. Every topic is covered twice — **our game first, then theirs** — so you can turn one page and see where a number came from. The tag in each heading says which game you are reading.

|                                                                                                                     | 8-BIT BUHURT | RETRO BOWL |
|---------------------------------------------------------------------------------------------------------------------|--------------|------------|
| The canvas and the screen                                                                                           | \$1          | \$2        |
| The fighter                                                                                                         | \$3          | \$4        |
| Colour                                                                                                              | \$5          | \$6        |
| The arena                                                                                                           | \$7          | \$8        |
| Portraits                                                                                                           | \$9          | \$10       |
| Type and the interface                                                                                              | \$11         | \$12       |
| \$13 the two side by side · \$14 the work order · \$15 the code · \$16 Retro Bowl's full sprite index · \$17 method |              |            |

If you are drawing, you need page 1, §3, §5 and §14. Everything else is why.

8-BIT BUHURT 1 The canvas and the screen

Four decisions were taken on 14 September 2026, from renders rather than opinion. This section is what an artist works from. Everything before it is the evidence.

## Locked.

- **The art grid is 480 × 270, drawn at ×2.** The engine keeps its 960 × 540 coordinate space; every asset is authored at half that and the renderer doubles it. One art pixel is two screen pixels. This is Retro Bowl's own trick — its match room is a 2600-wide room holding 1300-wide art at ×2 — and it is the single decision that makes this game look 8-bit rather than merely low-detail.
- **The list is 1.70 : 1.** 612 × 360 on screen, **306 × 180 authored.** Chosen from a four-way render against 1.90 (the old value), 1.50 and 1.40.
- **The list sits in a venue, not a void.** Far stand and fence above, side stands with the bench fighters at the rail, near rail and a row of crowd heads below. The old 106-px club banners and the 84-px card strip are cut to pay for it.
- **Fighters are sprites, recoloured by a key-colour shader** (\$2), not engine primitives.

Still open: whether the menus and HUD move onto the coarse grid too, or stay crisp at 960 × 540. That one is waiting on a render and does not block any of the art below.



The locked framing at 1.70 : 1. The floor and the fighters are the real Godot render at the new geometry; the stands, rails and crowd are blocked in to show the budget, not drawn as art. The HUD hint line and the two corner buttons still sit at their old anchors.

## The vertical budget

270 authored rows, every one accounted for. Authored on the left, what it becomes on screen on the right.

| BAND                              | AUTHORED        | ROWS       | ON SCREEN     |
|-----------------------------------|-----------------|------------|---------------|
| HUD — clock, formation, calls     | y 0-22          | 22         | 0-44          |
| Far stand — crowd, 2 frames       | y 22-40         | 18         | 44-80         |
| Far fence / barrier               | y 40-43         | 3          | 80-86         |
| <b>The list</b>                   | <b>y 43-223</b> | <b>180</b> | <b>86-446</b> |
| Near rail                         | y 223-226       | 3          | 446-452       |
| Near crowd — heads only, 2 frames | y 226-240       | 14         | 453-480       |
| Fighter card strip                | y 242-268       | 26         | 484-536       |

Horizontally the list runs **x 87 - 393** authored (174-822 on screen), leaving **87 authored px of side stand** on each side. The bench fighters stand at the rail inside that, crowd behind them.

**The redesign does not buy list height.** 184 authored rows today, 180 after. The bands added above and below eat exactly what the shrunk card strip frees. What it buys is that the 64 authored px of empty club banner down each side becomes 87 px of stand and bench, and the arena stops floating in a void. Worth knowing so nobody expects a bigger fight.

## RETRO BOWL 2 The canvas

Retro Bowl is a **480 × 270** game. Thirty-one of its thirty-three rooms are exactly that size, and so are their cameras. 480 × 270 is 1080p divided by four, which is why it scales to any modern screen with no filtering and no half-pixels.

| ROOM          | SIZE     | CAMERA  | WHAT IT IS                                                      |
|---------------|----------|---------|-----------------------------------------------------------------|
| 31 rooms      | 480×270  | 480×270 | every menu, roster, league, draft, hall of fame, options screen |
| rm_match      | 2600×540 | 960×540 | the only room you play in                                       |
| rm_kongregate | 960×540  | —       | portal splash, not gameplay                                     |

**The match room is not a second resolution.** It is the same 480 × 270 art grid in doubled coordinates. The field art is 1300 px wide and fills a 2600-wide room, so it is drawn at ×2; the camera is 960 × 540, so you are looking at **480 art-pixels of field at a time** — the same density as every menu. Working in 2× space buys sub-pixel movement for twenty-two running men without ever showing a half-pixel. The art never leaves the 480 × 270 grid.

The vertical budget is exact and worth copying: the field sprite is 237 px tall and the crowd strip above it is 33 px. 237 + 33 = 270. Nothing is left over and nothing is cropped.

The 1300×237 field, and the 480×270 window onto it



The camera pans along the length. Endzones (spr\_endzone, 100×169) and goal posts (spr\_goal\_post, 17×90) are drawn OVER the field, not painted into it - they carry club colour.

Screen geometry, to scale.

## 8-BIT BUHURT 3 The fighter

### The canvas, and what the sport allows

**72 × 72 authored, 144 × 144 on screen, for every pose and both classes.** Retro Bowl varies its canvas per pose — 16×16 standing, 16×24 jumping, 32×26 diving. One cell instead is cheaper in bookkeeping and costs only atlas space, which is free at this volume: one `cell`, one origin, and an artist who never has to ask how big this frame is.

**72 was measured, not chosen.** At **~18.9 authored px per metre** — 34 px of man for a 1.8 m fighter — a 120 cm polearm is 23 px and a 90 cm falchion is 17. The pose that sets the canvas is a **level polearm swing**: the head reaches about 32 px from the body centre, and an origin dead

centre means the canvas is twice that. 64 leaves 1 px of margin; 72 leaves 5. **Nothing but the feet may touch the cell edge** — if a pose does not fit, the pose is wrong, not the canvas.

**Read this before drawing a pose.** Buhurt rules, not preference (HMBIA / IMCF, via [medievalextreme.com](http://medievalextreme.com)):

- **No thrusts.** *“No stabs. Not with a sword, not with an axe, not with a polearm.”* Every strike in this set is a **swing**. This is the rule that sized the canvas; a thrust pose would be a drawn foul.
- **Illegal target zones:** neck, groin, back of the knee, spine, back of the head, feet. Do not show a swing or a throw landing on one.
- **A downed man is out.** Third point of contact — knee, hand or backside — and he does not get up. There is no get-up sheet on purpose: it is an animation that can never play, and the sim agrees ( `State.DOWN` holds for `DOWN_HOLD` , then `State.OUT` ).
- **No dual-wielding, no war hammers, no flails.** Punching with the shield *is* legal and is the natural second frame of `ss_swing` .

The sim has no weapon rules at all — `Act` is Bullrush / Clinch / Hit / Takedown / Hold / Escape / Break. **The two classes are a look, not a mechanic.**

Two things the engine draws, not the artist

**A 1 px near-black outline** around the whole silhouette is part of the sprite, and it is the one place this departs from Retro Bowl. They have no outline and do not need one: saturated kits on green grass. Ours are steel-grey men on brown earth standing 61 px apart, and two of them touching fuse without it. It is one of the twelve colours, not a thirteenth.

**A ground ring under every man**, 36 × 12 authored (72 × 24 on screen) — the drag target, and about a thumb at phone scale. Smaller and the man in front swallows it; at 36 the arcs clear him on both sides. Dim club-coloured rim when idle, route-yellow while the man is being dragged. Engine-drawn, not art.

The figure

Lane pitch is 180 authored rows ÷ 5 men = **36**. So:

|                             | AUTHORED       | ON SCREEN      | PIXELS PER FRAME |
|-----------------------------|----------------|----------------|------------------|
| Retro Bowl's player         | 16 × 17        | 32 × 34        | 272              |
| <b>Retro Buhurt fighter</b> | <b>24 × 36</b> | <b>48 × 72</b> | <b>864</b>       |
| The plan this replaces      | 48 × 80        | 48 × 80        | 3,840            |

## The same man on four grids — every one shown at its true on-screen size

|                                                                                   |                                                                                   |                                                                                     |                                                                                          |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|  |  |   |       |
| <b>Retro Bowl</b><br>16 x 17 authored<br>32 x 34 on screen<br>272 px per frame    | <b>PROPOSED</b><br>24 x 36 authored<br>48 x 72 on screen<br>864 px per frame      | <b>Bigger option</b><br>28 x 40 authored<br>56 x 80 on screen<br>1,120 px per frame | <b>Old plan, 1:1 grid</b><br>48 x 80 authored<br>48 x 80 on screen<br>3,840 px per frame |

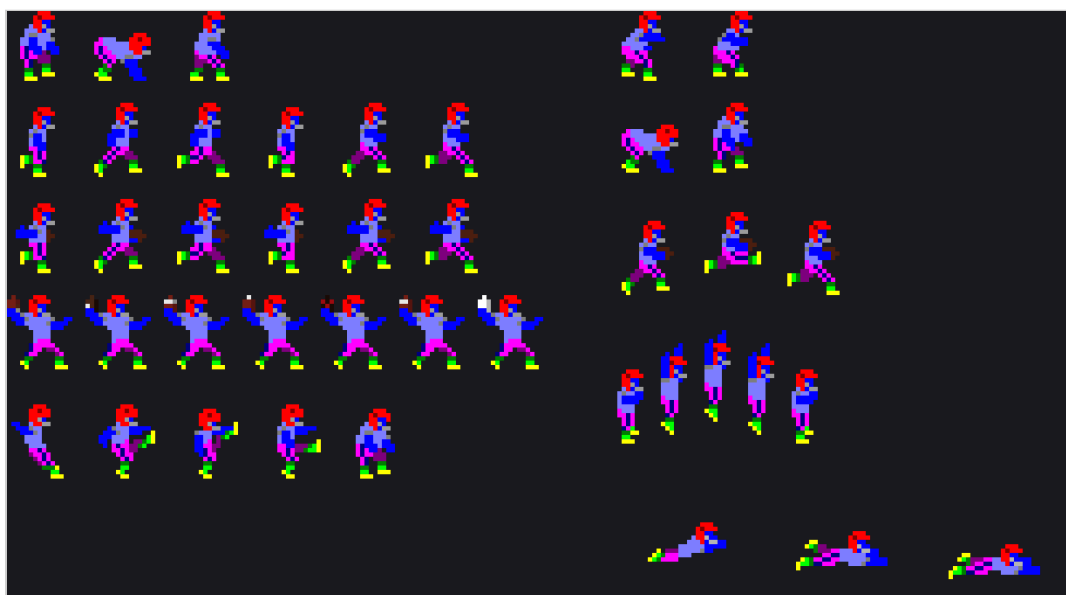
All four at their true on-screen size. The fighter blockouts are proportion studies, not art direction — they show the budget a helm, a divided surcoat and a mark have to fit into.

**The fighter is barely smaller on screen and 4.4× cheaper to draw.** 72 screen px tall against the old plan's 80, for 864 pixels a frame instead of 3,840. The grid decision does not shrink the man, it coarsens him. And 24 × 36 lands within four pixels of the **24 × 32** that REGISTER 05.3 measured by rendering as the size where the heraldry finally did any work — the only figure size in this project's history arrived at by looking rather than by arithmetic.

If more room to draw is wanted later there is exactly one lever: float the card strip over the list the way Retro Bowl floats its skip button. That reclaims 26 authored rows, giving a 206-row list, a 41-px lane and a **28 × 40** fighter.

## RETRO BOWL 4 The player

The single most surprising number in the build: **a Retro Bowl player is 16 × 17 pixels**, and the drawn figure inside that box is 8–13 px wide. The head is five pixels. At 270 px of screen height, a player is **6.3% of the screen**.



Every core animation at 9×, as stored. Left column: stand (3), run (6), stiff-arm (6), QB aim (7), place-kick (5). Right: wrestle (2), get-up (2), jump (3), jump-catch (5), dive (3).

## The animation table

Speeds are frames per second, straight out of the manifest. A speed of — means the frames are states the code indexes into, not an animation that plays.

| ACTION            | CANVAS | FRAMES | FPS   | CYCLE  | NOTE                                         |
|-------------------|--------|--------|-------|--------|----------------------------------------------|
| spr_stand         | 16×16  | 3      | —     | —      | three <i>poses</i> , picked by state         |
| spr_run           | 16×17  | 6      | 8     | 0.75 s | the workhorse cycle                          |
| spr_stiffarm      | 16×17  | 6      | 8     | 0.75 s | run cycle, arm out                           |
| spr_jockey        | 16×17  | 4      | 8     | 0.50 s | defender shuffle                             |
| spr_qb_walk       | 16×17  | 2      | 6     | 0.33 s | two frames is enough                         |
| spr_qb_aim        | 15×17  | 7      | 6     | 1.17 s | the most frames of any figure                |
| spr_qb_throw      | 12×17  | 1      | 1     | —      | <b>one frame</b> for the throw               |
| spr_wrestle       | 16×16  | 2      | 4     | 0.50 s | the tackle struggle — two frames             |
| spr_getup         | 16×16  | 2      | 4     | 0.50 s |                                              |
| spr_handoff       | 16×16  | 2      | 4     | 0.50 s |                                              |
| spr_falling       | 24×24  | 2      | 3     | 0.67 s | bigger box — he is horizontal                |
| spr_jump          | 16×24  | 3      | 8     | 0.38 s | taller box, same width                       |
| spr_jump_catch    | 7×28   | 5      | 8     | 0.63 s | tallest and narrowest — he is fully extended |
| spr_dive          | 32×26  | 3      | 8     | 0.38 s | widest box — full extension                  |
| spr_spike         | 16×20  | 3      | 9     | 0.33 s |                                              |
| spr_kick1 / kick2 | 16×20  | 3 / 5  | 8 / 9 | —      | punt and place-kick                          |
| spr_celebrate     | 16×20  | 2      | 2     | 1.0 s  | deliberately slow                            |

**The frame budget is the headline.** Thirty-two body sprites, **106 frames total**, cover every state a football player can be in. The run cycle gets six. Everything else gets two or three. A two-frame wrestle carries the most dramatic moment in the game. Nothing in Retro Bowl is animated for its own sake — frames are spent where the eye is already looking and nowhere else.

## The rules the manifest reveals

- **The canvas changes with the pose, and the origin holds the feet.** Standing poses are 16×16–17 with the origin at (7–8, 15–16) — bottom-centre. A jump grows the box *upward* to 16×24 and moves the origin to (7, 22), so the feet stay on the same ground line. A dive grows it *sideways* to 32×26. The engine never repositions the sprite; the canvas and origin do it.
- **There is one facing, mirrored.** No left-facing artwork exists in the build.
- **Ball-carrying is a parallel set, not an overlay.** `spr_run`, `spr_run_ball` and `spr_run_ball2` are three separate six-frame sets — carrying left, carrying right, and empty-handed. Three copies of the same cycle cost eighteen frames and remove every alignment problem an attached ball would have caused.
- **Loose balls are their own seven sprites.** `spr_ball1`...`spr_ball7`, each 10×5 with four frames at 10 fps — seven rotations of a tumbling ball, plus `spr_ball` (40×20, six frames) for the ball in flight and `spr_blur` for its trail.

### The fidelity — how 8-bit, exactly

Measured rather than asserted. Retro Bowl is **NES-sized and SNES-coloured**, and it keeps the two apart rigidly.

|                    | NES              | SNES      | RETRO BOWL                                   |
|--------------------|------------------|-----------|----------------------------------------------|
| Canvas             | 256 × 240        | 256 × 224 | <b>480 × 270</b>                             |
| Colours per sprite | 3 + transparent  | up to 16  | <b>11-13</b>                                 |
| Figure size        | 8×8 / 8×16 tiles | —         | <b>16 × 17</b> ( $\approx 2 \times 2$ tiles) |

Per-frame counts from the atlas: `spr_stand` 11-12, `spr_run` 12-13, `spr_wrestle` 12, `spr_dive` 11-12. The one outlier, `spr_qb_aim` at 14-16, is carrying a ball and a throwing arm.

It drops hard for anything that is not a person — helmet 7, cheerleader 8, coach 8, chains 6, goal post 4-6, loose ball **2** — and UI panels are **one colour**: `spr_box_208x80`, `spr_scoreboard` and `spr_star_rating` are single-colour silhouettes the game tints and writes over. The premium is spent entirely on **backgrounds**, where `spr_field` runs 41-212 colours a frame.

**Binary alpha on every figure.** Across all 632 frames, 31 sprites contain partial alpha and every one of them is UI or an effect — HUD panels, control hints, the clock, rain, motion blur. Of the eighteen player sprite families, **not one pixel is anything other than fully transparent or fully opaque**. That is not a style choice: the colour-replace shader tests `pixel.a == 1.0`, so a soft edge on a fighter renders as raw magenta.

**What carries the 8-bit read**, and all of it transfers: the figure at roughly two tiles, flat fills with no outline and no gradient, binary alpha, two to three frames per action with six for the one cycle that matters, and arcade-era type. Dithering only where there is room for it — they checker the 60 × 60 helmet's facemask and dither nothing on a 16 × 17 man.

### The budget for a 24 × 36 fighter: twelve colours, hard ceiling fourteen

Nine are the key slots above. Three are fixed and never swapped:

| #   | COLOUR        | JOB                                                                        |
|-----|---------------|----------------------------------------------------------------------------|
| 1-9 | the nine keys | steel, steel_b, helm, surcoat_a, surcoat_b, trim, leather, leather_b, mark |
| 10  | black         | visor slot, the deepest separation between overlapping men                 |
| 11  | near-black    | one step up — under the arms, behind the belt                              |
| 12  | white         | a single specular hit on the plate. One.                                   |

**Why NES-strict would break this game specifically.** Armour reads as metal through value steps. At three colours the harness eats the whole budget and the surcoat gets nothing — and the surcoat is the only thing that tells two grey men apart. REGISTER 05.3 measured that failure once already from the other direction, when the figure was too small for the heraldry to work. **The 8-bit is carried by size, flatness and frame count, not by a colour cap.**

### The key colours

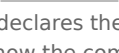
Nine slots, replacing Retro Bowl's eleven. Skin is dropped — the men are in helms — and what it frees goes to the heraldic division of the surcoat, which is what a club is actually recognised by. The three rules from §2 apply unchanged: **opaque pixels only, exact hex, never eyedropped, the shade is its own slot, not a darkening.**

| SLOT      | KEY                                                                               | WHAT IT PAINTS                                                 |
|-----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------|
| surcoat_a |  | #FF00FF surcoat field, primary tincture                        |
| surcoat_b |  | #800080 second tincture — per pale, per fess, quartered        |
| trim      |  | #FFFF00 edging, belt, straps                                   |
| helm      |  | #FF0000 helm cover, plume, crest                               |
| steel     |  | #8080FF plate — lets a club run blackened or polished          |
| steel_b   |  | #0000FF plate shadow                                           |
| leather   |  | #00FF00 gambeson, gauntlet cuffs, boots                        |
| leather_b |  | #008000 leather shadow                                         |
| mark      |  | #808080 a flat chest panel where IconBank stamps the club mark |

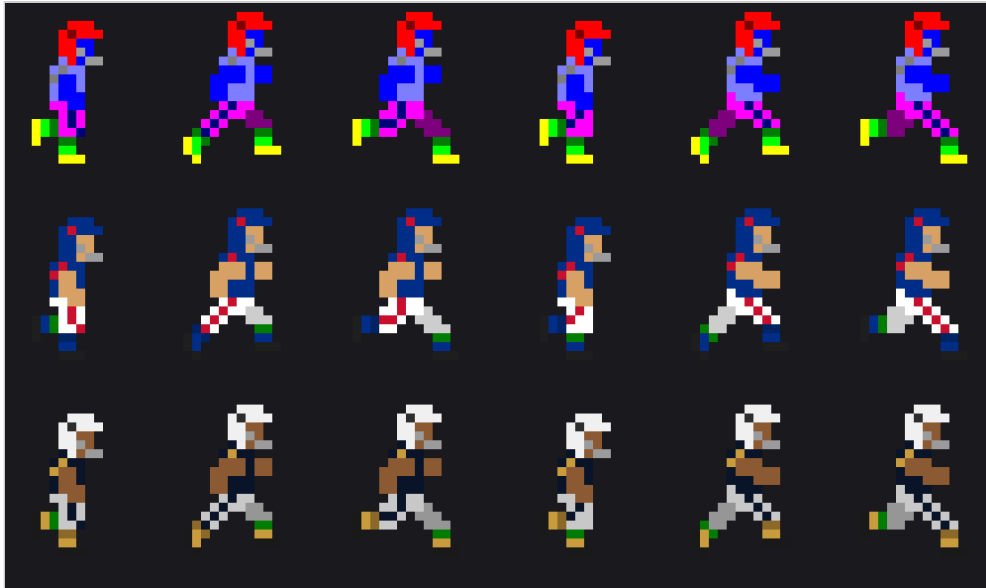
## RETRO BOWL 6 Colour

This is the part of Retro Bowl's art that is not obvious from playing it, and it is the part worth stealing. **The player sprites are not coloured artwork. They are masks.**

Every on-field figure is painted in eleven flat, arbitrary key colours — pure red, pure magenta, pure blue and so on. Nothing in the file is the colour you see in the game. At draw time a fragment shader, `sh_ColorReplaceBlend`, matches each pixel against the eleven keys and substitutes that team's kit colour for that slot. One set of sprites serves all thirty-two clubs, four kits each, home and away, for ever.

| SHADER UNIFORM | KEY                                                                                 | WHAT IT PAINTS                  |
|----------------|-------------------------------------------------------------------------------------|---------------------------------|
| colorHelmet    |  | #FF0000 helmet shell            |
| colorLogo      |  | #800000 helmet stripe / logo    |
| colorShirt     |  | #8080FF jersey body             |
| colorSleeve    |  | #808080 jersey sleeve           |
| colorPants     |  | #FF00FF pants                   |
| colorPants_b   |  | #800080 pants shade             |
| colorStripe    |  | #000080 pant stripe             |
| colorSocks     |  | #00FF00 socks upper             |
| colorSocks_b   |  | #008000 socks lower             |
| colorSkin      |  | #0000FF skin — arms, legs, face |
| colorCleats    |  | #FFFF00 cleats                  |

The eleven key colours, exactly as the shader declares them. Every one is a full-bright or half-bright primary — which is how the artist keeps them straight while drawing, and how the comparison stays exact.



The six-frame run cycle. Top: as it is stored in the atlas. Middle and bottom: the same six frames through a re-implementation of their shader with two different kits. One drawing, any club.

## The shader, in full

```
// sh_ColorReplaceBlend – fragment shader, verbatim from the build
uniform vec3 colorHelmet, colorLogo, colorShirt, colorSleeve, colorPants,
            colorPants_b, colorStripe, colorSocks, colorSocks_b,
            colorSkin, colorCleats;          // the eleven KEYS
uniform vec3 replaceHelmet, replaceLogo, ... replaceCleats; // this club's kit

void main() {
    vec4 pixel = texture2D( gm_BaseTexture, v_vTexcoord );
    if (pixel.a == 1.0) {
        vec3 eps = vec3(0.009, 0.009, 0.009);
        if (within(pixel, colorShirt, eps)) pixel = vec4(replaceShirt, 1.0);
        else if (within(pixel, colorSkin, eps)) pixel = vec4(replaceSkin, 1.0);
        else if (within(pixel, colorHelmet, eps)) pixel = vec4(replaceHelmet, 1.0);
        ... nine more, in a fixed priority order ...
    }
    gl_FragColor = pixel;
}
```

### Three consequences the artist has to know before drawing anything.

- **Only fully opaque pixels are recoloured.** The shader tests `pixel.a == 1.0`. Any anti-aliased edge, any soft shadow, any 50%-alpha pixel *keeps its key colour* and renders as raw magenta on screen. This is why Retro Bowl has no anti-aliasing anywhere: the palette system forbids it.
- **The match is exact to  $\pm 0.009$**  (about 2/255). A key colour that is one step off is not recoloured. Colours must be typed in, never eyedropped, never resampled, never saved through a lossy step.
- **Shading is a second slot, not a darker version of the first.** Pants and pants-shade are two separate uniforms; so are socks and socks-shade. The artist gets two tones per garment and the club decides what both of them are. There is no automatic darkening.

## What is not a key colour

Black, white, the facemask grey, and the brown of the ball are fixed — they are drawn once and never swapped. Notably there is **no outline colour at all**. Retro Bowl's figures have no black keyline; they read against the field purely on value contrast between the kit and the grass, which

is the reason the kit palettes are as saturated as they are.

The kit data

`uniforms_default.txt` ships one row per club, with eight colour slots per kit and four kits per club (home / away / alt / rush), plus three per-venue values. 32 clubs × 4 kits × 8 slots = **1,024 hand-picked colours**, in a tab-separated text file a designer can edit without opening the engine.

```
name tla home_helmet home_logo home_shirt home_sleeve home_pants home_stripe
home_socks1 home_socks2 [ ×4 kits ] roof rainfall snowfall endzone
Buffalo BUF $ffffff $003087 $003087 $c8102e $ffffff $c8102e $003087 $003087 ...
```

The last four columns are the reason two clubs' stadiums look different without any extra art: a **roof** flag, a **rainfall** and **snowfall** weighting, and an **endzone** colour. Weather is a property of the ground, not of the day.

8-BIT BUHURT 7 The arena

The list is **306 × 180 authored** (612 × 360 on screen) at 1.70 : 1, and it sits in a building rather than a void: a far stand and fence above, side stands with the bench fighters at the rail, a near rail and a row of crowd heads below. Five lanes run along the line, one man to each.

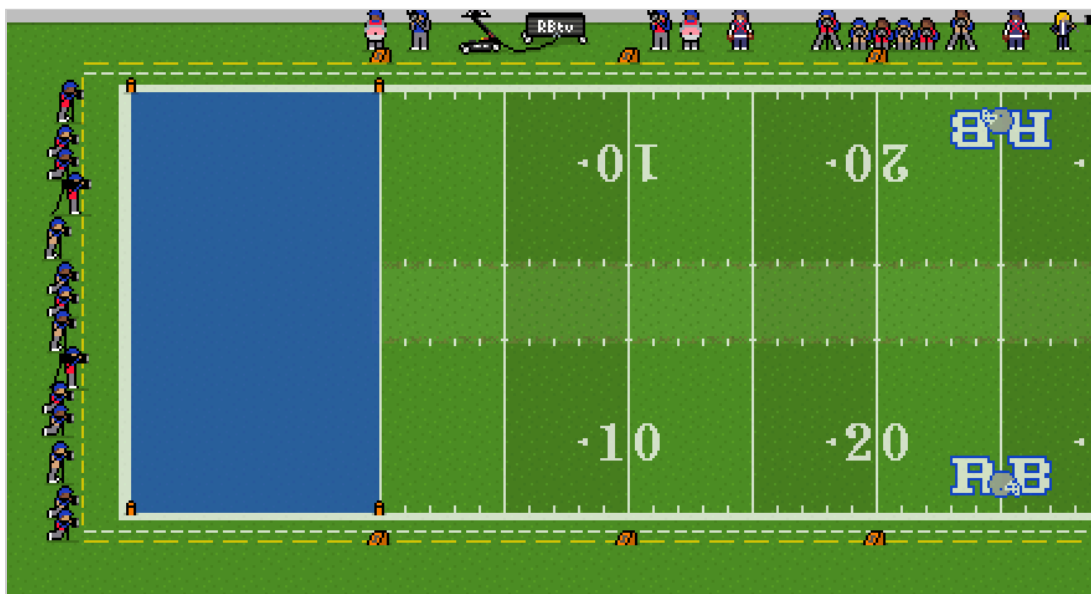
| ASSET                   | ON SCREEN | AUTHORED         | FRAMES |
|-------------------------|-----------|------------------|--------|
| List surface            | 612 × 360 | <b>306 × 180</b> | 2-4    |
| Far stand               | 960 × 36  | <b>480 × 18</b>  | 2      |
| Side stand (mirrored)   | 174 × 360 | <b>87 × 180</b>  | 2      |
| Near crowd, heads only  | 960 × 24  | <b>480 × 12</b>  | 2      |
| Rail / fence (tiles)    | 960 × 6   | <b>480 × 3</b>   | 1      |
| Arena grounds — UI art  | 520 × 300 | 520 × 300        | 6      |
| Venue splashes — UI art | 960 × 540 | 960 × 540        | 2      |

The surface is **whole variants, never a tint** — dry, wet, snow, indoor. It used to be specified greyscale so the game could multiply it by an occasion colour; that was dropped on 14 September and the greyscale rule went with it. Retro Bowl reached the same answer: seven fully coloured fields and no tint anywhere.

The crowd

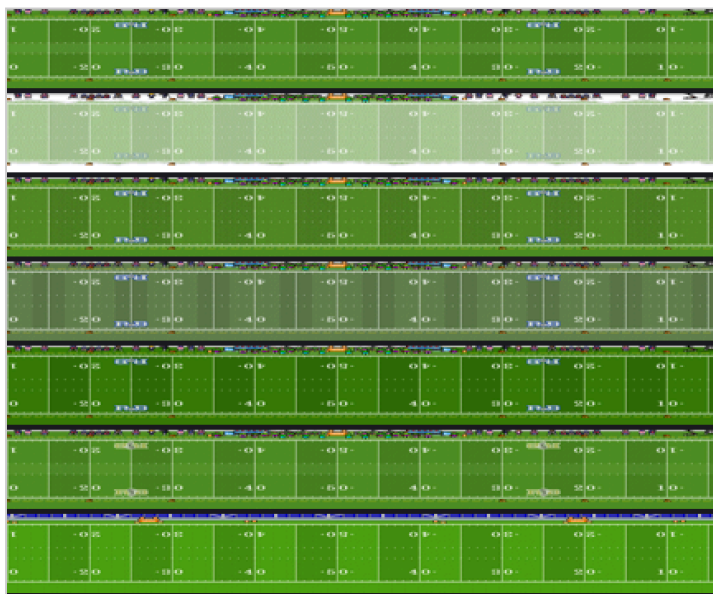
Retro Bowl's crowd strip is **21 colours, and the two most common are #262626 and #3B3B3B** — a dark mass with club-coloured specks and a few skin tones, not a rainbow. Two of the twenty-one are shader keys, so **the crowd wears the home club's kit**. Copy that exactly: it is why their stands recede instead of competing with the play.

A crowd figure is 3 × 4 authored pixels — a two-pixel head and a three-pixel body. Two frames per stand is the whole animation.



The left third of `spr_field` at 2x. Yard numbers, hash marks, sideline and the near crowd are all painted into the one image. The end zone is the flat keyed block — it is recoloured per club.

One  $1300 \times 237$  image holds the entire pitch. There is no tiling and no composition: the yard lines, the numbers, the hashes, the sideline benches and the bottom row of crowd are all baked in. Seven complete variants exist.



All seven `spr_field` frames, centre section. Reading down: standard grass, snow, a second grass dressing, worn/muddy, dark turf, a sand-patched variant, and a blue-sideline dome.

| ASSET                          | SIZE     | FRAMES | WHAT IT DOES                                                      |
|--------------------------------|----------|--------|-------------------------------------------------------------------|
| <code>spr_field</code>         | 1300×237 | 7      | the whole pitch, seven weather/surface dressings                  |
| <code>spr_fieldfans</code>     | 1300×33  | 2      | far crowd strip, two frames, 15 fps — the whole stand's animation |
| <code>spr_endzone</code>       | 100×169  | 7      | drawn over the field; takes the club's endzone colour             |
| <code>spr_goal_post</code>     | 17×90    | 3      | three states, origin at (0,85) — the base                         |
| <code>spr_sideline_crew</code> | 299×27   | 1      | one strip of staff, repeated along the touchline                  |
| <code>spr_scoreboard</code>    | 208×88   | 1      | <b>one colour</b> — a silhouette the game tints and writes over   |
| <code>spr_chains</code>        | 32×27    | 2      | the down markers                                                  |



Both frames of `spr_fieldfans`. A two-frame swap across a 1300-px strip is the entire crowd animation. Fans are 2-3 px each and read as a texture, not as people.

**The crowd is a texture, and that is the lesson.** A stand full of people costs two frames of a flat strip. Individual sideline figures — `spr_cheerleaders1..3` at 20×20 with ten frames each, `spr_coach` at 10×21 — exist only where a handful of them stand close to the camera. Everything at distance is pattern.



Sideline life and props at 6×: a cheerleader cycle, the group sprite, a coach, the chains, a goal post, a tumbling ball and the in-flight ball.

8-BIT BUHURT

9 Portraits

Twenty drawings, sixty-four men. Both layers share one **64 × 64** canvas with a transparent background: a **harness** drawn with the head area empty, and a **head** drawn with no armour on it. They meet at a gorget line four pixels deep, because a butt joint shows a seam at every scale the game draws these at.

| LAYER       | CANVAS  | COUNT | REGION ON THE SHARED CANVAS                                            |
|-------------|---------|-------|------------------------------------------------------------------------|
| Head + neck | 64 × 64 | 16    | y 6-38, about 26 px wide, centred. No armour, no shoulders, no collar. |
| Harness     | 64 × 64 | 4     | y 34-64, full width at the shoulders. No helmet, no face, no head.     |

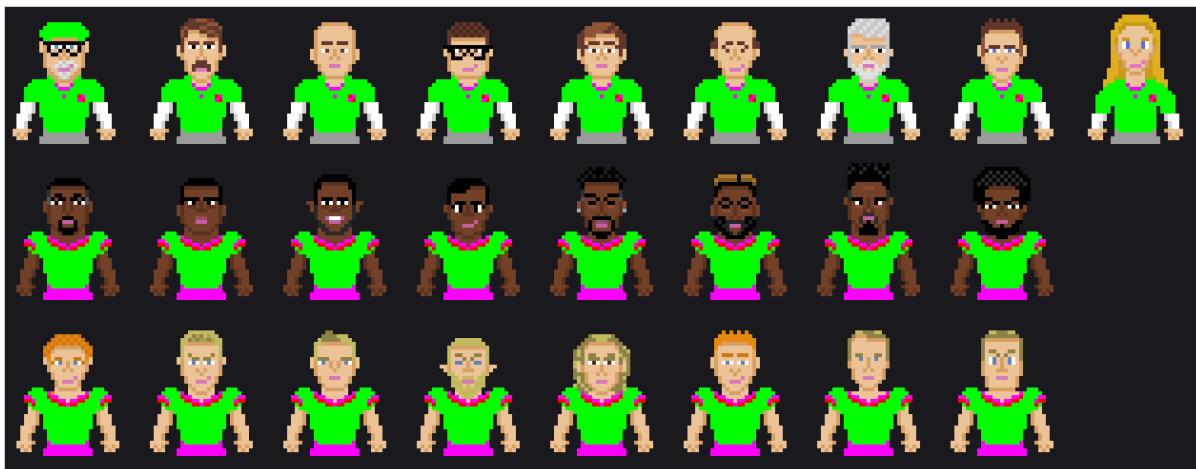
**These are UI art and stay at full density.** The fight screen is authored at half size and doubled; the menus are not, because the HUD and the type stayed at 960 × 540. A portrait sits next to type, so it keeps the type's resolution.

The four harnesses are not decoration — the game reads one off the fighter's rating, so a scruffy man in a kit-bash brigandine looks like a Backyard fighter and a man in Milanese plate looks like he belongs at Worlds. The player reads a squad's quality off the cards before he reads a number.

**This is the one place we beat them outright.** Retro Bowl draws **132 separate portrait frames** to cover its faces. Four harnesses × sixteen heads is **64 men from 20 drawings**, and it scales multiplicatively if either layer grows. Keep it.

## RETRO BOWL 10 Portraits

Retro Bowl spends more art on faces than on football. The portrait bank is **sixteen sprites at 32 × 36, 132 frames in total** — more frames than the entire on-field figure set, for a thing that appears on menu screens.



Three of the sixteen portrait sheets at 5×. Each is one skin tone with eight or nine head variants; the shirt is key-coloured and takes the club's colour, so one sheet dresses every team.

| BANK               | SHEETS | PER SHEET | TOTAL | WHAT IT IS         |
|--------------------|--------|-----------|-------|--------------------|
| spr_coaches_01..04 | 4      | 9         | 36    | staff faces        |
| spr_dark_001..008  | 8      | 8         | 64    | darker skin tones  |
| spr_light_001..003 | 3      | 8         | 24    | lighter skin tones |
| spr_mix_001        | 1      | 8         | 8     | mixed              |

All sixteen share one canvas and one crop — head to mid-thigh, arms down, dead centre, neutral expression — so any face can appear in any frame with no layout code. The shirt and a small chest accent are keyed, which is what makes 132 drawings cover every player on every roster.



`spr_helmet`, 60×60, two frames — with and without a centre stripe. Shell, stripe and trim are three separate keys, so the team-select screen shows a real helmet in real club colours from one drawing.

## 8-BIT BUHURT 11 Type and the interface

**The UI did not follow the field onto the coarse grid.** Pete, 14 September: “*HUD should stay at the 960 × 540, it has 8 bit font to match.*” World art is authored at half size and doubled; menus, the HUD and their art are drawn at full density. That split is standard practice and it is what keeps type readable on a phone.

Four faces, all drawn for this game or derived for it, and **all declared on an 8 px em**:

| FACE          | ROLE                                         | ORIGIN                                                                           |
|---------------|----------------------------------------------|----------------------------------------------------------------------------------|
| Buhurt Rail   | body — anything small and dense              | ours                                                                             |
| Buhurt Plate  | headings, names, numbers (monospaced digits) | Press Start 2P, respaced proportional. SIL OFL, renamed as the licence requires. |
| Buhurt Gorget | unused, held                                 | ours                                                                             |
| Buhurt Maul   | unused, held                                 | ours                                                                             |

**A pixel face is crisp only at whole multiples of its em.** The ladder is **8 / 16 / 24 / 32** and **nothing between**, and `Ui.GRID` says so. But `Ui.snap()` — the function the comment names as the enforcer — **has no caller on the text path**. The real call sites are 85 at size 13, 64 at 12, 41 at 14 and 34 at 11: an 8 px em rendered at  $\times 1.375$ ,  $\times 1.5$ ,  $\times 1.625$  and  $\times 1.75$ .

At 16 / 24 / 32 the glyph pixel is 2, 3 or 4 screen pixels, so the type agrees with the  $\times 2$  world grid and the two layers look like one game. At 11–14 it agrees with nothing. Fixing it moves most body text 13  $\rightarrow$  16 and reflows the dense screens, so it is listed as work rather than done.

## RETRO BOWL 12 Type and the interface

The interface is built from fixed-size panels rather than nine-slices. Every panel exists at the exact pixel size it is used at, all of them multiples of 16 on a  $480 \times 270$  grid:  $64 \times 32$ ,  $64 \times 64$ ,  $64 \times 80$ ,  $128 \times 128$ ,  $160 \times 128$ ,  $160 \times 160$ ,  $176 \times 176$ ,  $208 \times 32$ ,  $208 \times 64$ ,  $208 \times 80$ ,  $208 \times 176$ ,  $224 \times 176$ ,  $288 \times 160$ ,  $416 \times 144$ ,  $448 \times 80$ . Fifteen panels, one frame each. Buttons come in ten fixed widths, every one 34 px tall with three frames — up, down, disabled.

Star ratings are a single sprite with eleven frames: `spr_star_rating`,  $39 \times 7$ , zero to five stars in half-star steps, in six flavours (small/big, left/centre aligned, with and without a number beside it). A rating is one draw call, never five.

### The five fonts, by name

| FONT                         | TYPEFACE           | SIZE    | CELL              | USED FOR                            |
|------------------------------|--------------------|---------|-------------------|-------------------------------------|
| <code>fnt_tecmo</code>       | Emulogic           | 11      | $15 \times 19$    | the body font — monospaced, NES-era |
| <code>fnt_tecmo_small</code> | 8-bit Operator+ 8  | 9 bold  | prop. $\times 14$ | dense lists and tables              |
| <code>fnt_tecmo_big</code>   | 8-bit Operator+ SC | 16 bold | prop. $\times 30$ | headings and the scoreboard         |
| <code>fnt_restart</code>     | Press Start 2P     | 6       | $8 \times 8$      | small caps chrome                   |
| <code>fnt_restart_big</code> | Press Start 2P     | 11      | $15 \times 15$    | large chrome                        |

All five are pre-rendered to the atlas with **anti-aliasing off**, which is consistent with the palette rule in §2. Press Start 2P is SIL Open Font License; Emulogic and 8-bit Operator+ are freely distributed. None of them is bespoke.

## 13 The two, side by side

| MEASURE                   | 8-BIT BUHURT                        | RETRO BOWL                                    |
|---------------------------|-------------------------------------|-----------------------------------------------|
| Art grid                  | $480 \times 270$ , drawn $\times 2$ | $480 \times 270$                              |
| Screen                    | $960 \times 540$                    | $480 \times 270$ ( $960 \times 540$ in match) |
| Playing surface, authored | $306 \times 180$                    | $1300 \times 237$                             |
| Surface visible at once   | all of it — 1.70 : 1                | a 480-px slice — 2.03 : 1                     |

|                          |                            |                               |
|--------------------------|----------------------------|-------------------------------|
| Bodies on screen         | 10                         | up to 22                      |
| Figure canvas            | 72 × 72 (one cell)         | 16 × 17 to 32 × 26 (per pose) |
| Figure itself            | 24 × 36                    | ~10 × 17                      |
| Figure frames            | <b>52</b> — 26 × 2 classes | <b>106</b> — 32 sprites       |
| Colours per figure frame | 12, ceiling 14             | 11–13                         |
| Runtime colour keys      | 9                          | 11                            |
| Outline on the figure    | 1 px near-black            | none                          |
| Portrait canvas          | 64 × 64, two layers        | 32 × 36, flat                 |
| Portrait drawings        | <b>20</b> → 64 men         | <b>132</b> frames             |
| Crowd animation          | 2 frames per stand         | 2 frames, 1300 × 33           |
| Type                     | 4 faces, 8 px em           | 5 faces, all third-party      |
| Engine                   | Godot 4.6                  | GameMaker Studio 2            |

The two columns are close on every line that matters and deliberately apart on three. We spend **half their figure frames** because there is no ball to carry and no second facing to draw. We spend **a sixth of their portrait drawings** because ours composite. And we carry an outline they do not need, because steel-grey men on brown earth fuse where saturated kits on green grass do not.

## 8-BIT BUHURT 14 The work order

Every canvas below is the authored size. The renderer doubles it. Nothing is drawn with anti-aliasing, a soft edge, or a partial-alpha pixel.

| ASSET                                                                                                                                     | CANVAS   | FRAMES | NOTE                  |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|-----------------------|
| <b>The fighter</b> — two weapon classes, one facing, mirrored. <b>Every frame on one 72 × 72 cell</b> , origin dead centre at the bottom. |          |        |                       |
| art/sprite/ss_idle.png                                                                                                                    | 216 × 72 | 3      | falchion/axe + shield |
| art/sprite/ss_walk.png                                                                                                                    | 288 × 72 | 4      | falchion/axe + shield |
| art/sprite/ss_charge.png                                                                                                                  | 432 × 72 | 6      | falchion/axe + shield |
| art/sprite/ss_swing.png                                                                                                                   | 288 × 72 | 4      | falchion/axe + shield |
| art/sprite/ss_clinch.png                                                                                                                  | 144 × 72 | 2      | falchion/axe + shield |
| art/sprite/ss_takedown.png                                                                                                                | 216 × 72 | 3      | falchion/axe + shield |
| art/sprite/ss_recover.png                                                                                                                 | 144 × 72 | 2      | falchion/axe + shield |
| art/sprite/ss_down.png                                                                                                                    | 144 × 72 | 2      | falchion/axe + shield |
| art/sprite/pa_idle.png                                                                                                                    | 216 × 72 | 3      | polearm               |
| art/sprite/pa_walk.png                                                                                                                    | 288 × 72 | 4      | polearm               |
| art/sprite/pa_charge.png                                                                                                                  | 432 × 72 | 6      | polearm               |
| art/sprite/pa_swing.png                                                                                                                   | 288 × 72 | 4      | polearm               |
| art/sprite/pa_clinch.png                                                                                                                  | 144 × 72 | 2      | polearm               |
| art/sprite/pa_takedown.png                                                                                                                | 216 × 72 | 3      | polearm               |
| art/sprite/pa_recover.png                                                                                                                 | 144 × 72 | 2      | polearm               |
| art/sprite/pa_down.png                                                                                                                    | 144 × 72 | 2      | polearm               |
| <b>52 frames — 26 a class.</b> The body is 24 × 36 of the cell; the weapon uses the rest.                                                 |          |        |                       |
| <b>The venue</b>                                                                                                                          |          |        |                       |

|                                                   |           |     |                                                               |
|---------------------------------------------------|-----------|-----|---------------------------------------------------------------|
| List surface                                      | 306 × 180 | 2–4 | whole variants (dry, wet, snow, indoor) — <b>never a tint</b> |
| Far stand                                         | 480 × 18  | 2   | full width, two-frame swap                                    |
| Side stand                                        | 87 × 180  | 2   | one drawing, mirrored                                         |
| Near crowd, heads                                 | 480 × 14  | 2   |                                                               |
| Rail / fence                                      | 480 × 3   | 1   | tiles horizontally                                            |
| <b>Everything else, halved from the old brief</b> |           |     |                                                               |
| Arena grounds                                     | 260 × 150 | 6   | was 520×300                                                   |
| Venue splashes (away, neutral)                    | 480 × 270 | 2   | was 960×540                                                   |
| Portrait harnesses                                | 32 × 32   | 4   | was 64×64 — now Retro Bowl's own portrait size                |
| Portrait heads                                    | 32 × 32   | 16  | same shared canvas, gorget line at y 19                       |

The two-layer portrait scheme survives the halving and should not be given up: four harnesses × sixteen heads is **64 men from 20 drawings**, against the 132 separate frames Retro Bowl draws for the same job.

## 8-BIT BUHURT 15 What has to change in the code

1. `Tuning.LIST_H` 570 → **510**. This is the one constant that invalidates the whole melee suite; everything is re-measured after it moves, once, deliberately.
2. `melee_scene.LIST_SCALE` 1.23 → **1.20**, `LIST_ORIGIN.y` 58 → **86**.
3. Banners out; far / side / near stand bands in. Card strip `CARD_H` 84 → 52, `STRIP_Y` 436 → 484. The two corner buttons need re-anchoring.
4. A fighter-sprite slot in `ArtBank.SLOTS` — there is none today, and `test_arena.gd` asserts the table and `ART.md` agree, so art can be finished with nowhere to land.
5. **Delete the greyscale rule** from `ART.md` and `ArtBank`. It exists to serve the mood tint, which was dropped on 14 September. Retro Bowl ships seven coloured field variants and tints nothing; variants is the answer.
6. Halve every `size` in `ArtBank.SLOTS` and draw at ×2.
7. A `shader_type canvas_item` port of §2 — nine equality tests, opaque pixels only, ±0.009.

## 16 Appendix — Retro Bowl's full sprite index

All 205 sprites, all 632 frames, straight from the manifest. Canvas is the authored PNG size; origin is the draw anchor in pixels from the top-left; fps is the manifest playback speed, and — means the frames are indexed states rather than an animation.

### Body — the on-field figure 32 sprites · 106 frames

| SPRITE               | CANVAS | FRAMES | FPS | ORIGIN |
|----------------------|--------|--------|-----|--------|
| spr_stand            | 16×16  | 3      | —   | 8,15   |
| spr_stand_ball       | 16×16  | 3      | —   | 7,15   |
| spr_stand_ball2      | 16×16  | 3      | —   | 7,15   |
| spr_stand_highlight  | 16×24  | 3      | —   | 8,22   |
| spr_run              | 16×17  | 6      | 8   | 7,16   |
| spr_run_ball         | 16×16  | 6      | 8   | 7,15   |
| spr_run_ball2        | 16×16  | 6      | 8   | 7,15   |
| spr_stiffarm         | 16×17  | 6      | 8   | 7,15   |
| spr_stiffarm2        | 16×17  | 6      | 8   | 7,15   |
| spr_wrestle          | 16×16  | 2      | 4   | 9,15   |
| spr_getup            | 16×16  | 2      | 4   | 7,15   |
| spr_getup_ball       | 16×16  | 2      | 4   | 7,15   |
| spr_getup_ball2      | 16×16  | 2      | 4   | 7,15   |
| spr_jump             | 16×24  | 3      | 8   | 7,22   |
| spr_jump2            | 16×24  | 3      | 8   | 7,22   |
| spr_jump_catch       | 7×28   | 5      | 8   | 3,27   |
| spr_jump_catch_ball  | 7×28   | 5      | 8   | 3,27   |
| spr_jump_catch_ball2 | 7×28   | 5      | 8   | 3,27   |
| spr_dive             | 32×26  | 3      | 8   | 16,25  |
| spr_dive_ball        | 32×26  | 3      | 8   | 16,25  |
| spr_dive_ball2       | 32×26  | 3      | 8   | 8,17   |
| spr_falling          | 24×24  | 2      | 3   | 12,22  |
| spr_falling_ball     | 24×24  | 2      | 3   | 12,22  |
| spr_falling_ball2    | 24×24  | 2      | 3   | 12,22  |
| spr_spike            | 16×20  | 3      | 9   | 8,19   |
| spr_spike2           | 16×20  | 3      | 9   | 8,19   |
| spr_handoff          | 16×16  | 2      | 4   | 8,15   |
| spr_celebrate        | 16×20  | 2      | 2   | 8,19   |
| spr_dummy            | 16×16  | 3      | —   | 7,13   |
| spr_jockey           | 16×17  | 4      | 8   | 7,15   |
| spr_place            | 16×13  | 2      | —   | 6,11   |
| spr_highlight        | 16×16  | 1      | 15  | 8,7    |

### Quarterback and kicker 9 sprites · 22 frames

| SPRITE | CANVAS | FRAMES | FPS | ORIGIN |
|--------|--------|--------|-----|--------|
|--------|--------|--------|-----|--------|

|                  |       |   |    |      |
|------------------|-------|---|----|------|
| spr_qb_walk      | 16×17 | 2 | 6  | 8,16 |
| spr_qb_aim       | 15×17 | 7 | 6  | 7,16 |
| spr_qb_aim_old   | 15×17 | 1 | 6  | 7,16 |
| spr_qb_throw     | 12×17 | 1 | 1  | 6,16 |
| spr_kick1        | 16×20 | 3 | 8  | 8,17 |
| spr_kick2        | 16×20 | 5 | 9  | 8,17 |
| spr_kickpoint    | 25×7  | 1 | 15 | 6,3  |
| spr_aim          | 7×7   | 1 | 15 | 2,3  |
| spr_route_pointR | 16×16 | 1 | 15 | 8,7  |

## The ball 10 sprites · 36 frames

| SPRITE         | CANVAS | FRAMES | FPS | ORIGIN |
|----------------|--------|--------|-----|--------|
| spr_ball       | 40×20  | 6      | —   | 20,10  |
| spr_ball1      | 10×5   | 4      | 10  | 5,2    |
| spr_ball2      | 10×5   | 4      | 10  | 5,2    |
| spr_ball3      | 10×5   | 4      | 10  | 5,2    |
| spr_ball4      | 10×5   | 4      | 10  | 5,2    |
| spr_ball5      | 10×5   | 4      | 10  | 5,2    |
| spr_ball6      | 10×5   | 4      | 10  | 5,2    |
| spr_ball7      | 10×5   | 4      | 10  | 5,2    |
| spr_blur       | 10×5   | 1      | 7   | 5,1    |
| spr_bouncespot | 16×16  | 1      | 15  | 4,3    |

## Field and venue 12 sprites · 29 frames

| SPRITE            | CANVAS   | FRAMES | FPS | ORIGIN |
|-------------------|----------|--------|-----|--------|
| spr_field         | 1300×237 | 7      | —   | 0,0    |
| spr_fieldfans     | 1300×33  | 2      | 15  | 0,0    |
| spr_endzone       | 100×169  | 7      | 15  | 0,0    |
| spr_goal_post     | 17×90    | 3      | —   | 0,85   |
| spr_goal_stand    | 10×37    | 2      | —   | 0,32   |
| spr_goal_shadow   | 32×35    | 1      | —   | 3,17   |
| spr_chains        | 32×27    | 2      | 15  | 16,26  |
| spr_sideline_crew | 299×27   | 1      | 15  | 0,0    |
| spr_grass         | 32×32    | 1      | 15  | 0,0    |
| spr_cone          | 7×4      | 1      | 15  | 3,2    |
| spr_scoreboard    | 208×88   | 1      | 15  | 0,0    |
| spr_matchline_dot | 32×16    | 1      | 15  | 16,8   |

## Sideline life 10 sprites · 39 frames

| SPRITE            | CANVAS | FRAMES | FPS | ORIGIN |
|-------------------|--------|--------|-----|--------|
| spr_cheerleaders  | 100×25 | 2      | 2   | 50,24  |
| spr_cheerleaders1 | 20×20  | 10     | 4   | 10,20  |
| spr_cheerleaders2 | 20×20  | 10     | 4   | 10,20  |

|                      |        |    |    |       |
|----------------------|--------|----|----|-------|
| spr_cheerleaders3    | 20×20  | 10 | 4  | 10,20 |
| spr_coach            | 10×21  | 1  | 15 | 5,19  |
| spr_fans             | 12×14  | 1  | 15 | 6,7   |
| spr_team_celebration | 100×40 | 1  | 15 | 0,0   |
| spr_press_conf       | 100×40 | 1  | 15 | 0,0   |
| spr_commentary_bg    | 16×16  | 2  | —  | 0,0   |
| spr_spangle          | 23×7   | 1  | 15 | 11,4  |

## Weather and effects 7 sprites · 15 frames

| SPRITE        | CANVAS | FRAMES | FPS | ORIGIN |
|---------------|--------|--------|-----|--------|
| spr_rain      | 2×8    | 2      | 8   | 0,7    |
| spr_snow      | 3×3    | 1      | 9   | 1,1    |
| spr_splash    | 13×12  | 5      | 3   | 6,9    |
| spr_scanlines | 64×60  | 1      | 15  | 0,0    |
| spr_square    | 16×16  | 1      | 15  | 0,0    |
| spr_box       | 4×4    | 4      | 15  | 0,0    |
| spr_wind      | 22×21  | 1      | 15  | 9,10   |

## Portraits and the face bank 19 sprites · 137 frames

| SPRITE          | CANVAS | FRAMES | FPS | ORIGIN |
|-----------------|--------|--------|-----|--------|
| spr_coaches_01  | 32×36  | 9      | 15  | 0,0    |
| spr_coaches_02  | 32×36  | 9      | 15  | 0,0    |
| spr_coaches_03  | 32×36  | 9      | 15  | 0,0    |
| spr_coaches_04  | 32×36  | 9      | 15  | 0,0    |
| spr_dark_001    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_002    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_003    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_004    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_005    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_006    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_007    | 32×36  | 8      | 15  | 0,0    |
| spr_dark_008    | 32×36  | 8      | 15  | 0,0    |
| spr_light_001   | 32×36  | 8      | 15  | 0,0    |
| spr_light_002   | 32×36  | 8      | 15  | 0,0    |
| spr_light_003   | 32×36  | 8      | 15  | 0,0    |
| spr_mix_001     | 32×36  | 8      | 15  | 0,0    |
| spr_helmet      | 60×60  | 2      | —   | 30,30  |
| spr_jacket      | 32×32  | 1      | 15  | 0,0    |
| spr_player_edit | 15×20  | 2      | —   | 7,10   |

## Ratings, meters and trophies 31 sprites · 122 frames

| SPRITE   | CANVAS | FRAMES | FPS | ORIGIN |
|----------|--------|--------|-----|--------|
| spr_star | 7×7    | 1      | 15  | 3,4    |

|                     |        |    |    |       |
|---------------------|--------|----|----|-------|
| spr_star_b          | 9×9    | 1  | 15 | 4,4   |
| spr_star_s          | 5×4    | 1  | 15 | 2,2   |
| spr_star_rating     | 39×7   | 11 | 15 | 0,3   |
| spr_star_rating_b   | 41×9   | 11 | 15 | 0,4   |
| spr_star_rating_bn  | 41×9   | 11 | 15 | 19,3  |
| spr_star_rating_c   | 39×7   | 11 | 15 | 19,3  |
| spr_star_rating_cb  | 41×9   | 11 | 15 | 20,4  |
| spr_star_rating_cbn | 41×9   | 11 | 15 | 20,4  |
| spr_attitude        | 11×11  | 7  | 15 | 5,5   |
| spr_attitude_s      | 5×5    | 7  | 15 | 2,2   |
| spr_condition       | 11×11  | 4  | 15 | 5,5   |
| spr_morale          | 9×9    | 1  | 15 | 4,4   |
| spr_workrate        | 96×32  | 3  | —  | 0,0   |
| spr_meter           | 192×20 | 1  | 15 | 0,0   |
| spr_xp              | 14×7   | 1  | 15 | 7,3   |
| spr_credit          | 9×9    | 6  | —  | 4,4   |
| spr_credit2         | 14×9   | 1  | —  | 6,4   |
| spr_heart           | 9×9    | 1  | 15 | 4,5   |
| spr_rookie          | 11×11  | 1  | —  | 5,5   |
| spr_trade           | 11×11  | 1  | 15 | 5,5   |
| spr_medals          | 7×12   | 4  | 15 | 0,0   |
| spr_medals_c        | 7×12   | 4  | 15 | 3,5   |
| spr_sb_ring         | 11×11  | 1  | 15 | 0,0   |
| spr_sb_ring_c       | 11×11  | 1  | 15 | 5,5   |
| spr_cup             | 15×15  | 1  | 15 | 7,7   |
| spr_trophy          | 32×32  | 1  | 15 | 15,15 |
| spr_hof             | 32×32  | 2  | —  | 0,0   |
| spr_hof_btn         | 32×32  | 2  | —  | 16,16 |
| spr_hof_c           | 32×32  | 2  | —  | 16,16 |
| spr_hof_s           | 7×8    | 1  | 15 | 3,0   |

## Panels, buttons and chrome 75 sprites · 126 frames

| SPRITE          | CANVAS  | FRAMES | FPS | ORIGIN |
|-----------------|---------|--------|-----|--------|
| spr_adRG        | 480×270 | 1      | 30  | 0,0    |
| spr_app_apple   | 110×32  | 1      | 15  | 0,0    |
| spr_app_discord | 110×32  | 1      | —   | 0,0    |
| spr_app_google  | 110×32  | 1      | 15  | 0,0    |
| spr_app_reddit  | 110×32  | 1      | 15  | 0,0    |
| spr_app_twitter | 110×32  | 1      | —   | 0,0    |
| spr_arrowL      | 32×32   | 1      | 15  | 16,16  |
| spr_arrowL2     | 8×15    | 1      | 15  | 7,7    |
| spr_arrowR      | 32×32   | 1      | 15  | 16,16  |

|                      |         |   |    |         |
|----------------------|---------|---|----|---------|
| spr_arrowR2          | 8×15    | 1 | 15 | 2,7     |
| spr_arrowR3          | 6×13    | 1 | 15 | 2,6     |
| spr_arrowU           | 32×32   | 1 | 15 | 16,16   |
| spr_audible          | 96×32   | 5 | —  | 0,0     |
| spr_box_128x128      | 128×128 | 1 | 15 | 0,0     |
| spr_box_160x128      | 160×128 | 1 | 15 | 0,0     |
| spr_box_160x160      | 160×160 | 1 | 15 | 0,0     |
| spr_box_176x176      | 176×176 | 1 | 15 | 0,0     |
| spr_box_208x176      | 208×176 | 1 | 15 | 0,0     |
| spr_box_208x32       | 208×32  | 1 | 15 | 0,0     |
| spr_box_208x64       | 208×64  | 1 | 15 | 0,0     |
| spr_box_208x80       | 208×80  | 1 | 15 | 0,0     |
| spr_box_224x176      | 224×176 | 1 | 15 | 0,0     |
| spr_box_288x160      | 288×160 | 1 | 15 | 0,0     |
| spr_box_416x144      | 416×144 | 1 | 15 | 0,0     |
| spr_box_416x144_fill | 416×144 | 1 | 15 | 0,0     |
| spr_box_448x80       | 448×80  | 1 | 15 | 0,0     |
| spr_box_64x32        | 64×32   | 1 | 15 | 0,0     |
| spr_box_64x41        | 64×41   | 1 | 15 | 0,0     |
| spr_box_64x64        | 64×64   | 1 | 15 | 0,0     |
| spr_box_64x80        | 64×80   | 1 | 15 | 0,0     |
| spr_button_104       | 106×34  | 3 | —  | 0,0     |
| spr_button_32        | 33×34   | 3 | —  | 0,0     |
| spr_button_32_9      | 33×34   | 3 | —  | 0,0     |
| spr_button_40        | 41×34   | 3 | —  | 0,0     |
| spr_button_50        | 54×34   | 3 | —  | 0,0     |
| spr_button_64        | 68×34   | 3 | —  | 0,0     |
| spr_button_74        | 74×34   | 3 | —  | 0,0     |
| spr_button_74s       | 74×26   | 3 | 15 | 0,0     |
| spr_button_90        | 90×34   | 3 | —  | 0,0     |
| spr_button_thin      | 41×18   | 3 | —  | 0,0     |
| spr_checkbox         | 16×16   | 2 | —  | 0,0     |
| spr_checkbox012      | 16×16   | 3 | —  | 0,0     |
| spr_clock            | 96×32   | 3 | 15 | 0,0     |
| spr_clock2           | 96×32   | 2 | 15 | 0,0     |
| spr_col_picker       | 391×262 | 1 | 15 | 195,131 |
| spr_connect          | 7×7     | 1 | 15 | 3,3     |
| spr_controls         | 140×59  | 1 | 15 | 70,29   |
| spr_controls_handoff | 140×59  | 1 | 15 | 70,29   |
| spr_controls_kick    | 140×59  | 1 | 15 | 70,29   |
| spr_controls_run     | 140×59  | 1 | 15 | 70,29   |
| spr_controls_runb    | 140×59  | 1 | 15 | 70,29   |

|                       |         |   |    |         |
|-----------------------|---------|---|----|---------|
| spr_controls_scramble | 140×59  | 1 | 15 | 70,29   |
| spr_controls_throw    | 140×59  | 1 | 15 | 70,29   |
| spr_cross             | 32×32   | 1 | 15 | 15,16   |
| spr_details           | 32×32   | 1 | 15 | 16,16   |
| spr_edit              | 14×14   | 1 | 15 | 7,7     |
| spr_fieldgoalbtn      | 96×32   | 1 | —  | 0,0     |
| spr_firstdown         | 20×20   | 7 | 4  | 10,19   |
| spr_gamepad_any       | 20×20   | 4 | 15 | 12,7    |
| spr_gamepad_xbox1     | 20×20   | 4 | 15 | 12,7    |
| spr_home              | 32×32   | 1 | 15 | 15,15   |
| spr_info              | 16×16   | 1 | 15 | 0,0     |
| spr_msg_bg            | 64×64   | 1 | 15 | 0,0     |
| spr_pointer           | 9×12    | 2 | —  | 4,6     |
| spr_profile           | 32×32   | 1 | 15 | 16,16   |
| spr_spanner           | 32×32   | 1 | 15 | 15,15   |
| spr_store             | 15×15   | 1 | 15 | 7,7     |
| spr_textbox48         | 48×16   | 1 | 15 | 24,8    |
| spr_textbox64         | 64×16   | 1 | 15 | 32,8    |
| spr_tick              | 32×32   | 1 | 15 | 16,16   |
| spr_training          | 16×23   | 1 | 15 | 8,11    |
| spr_uniform_select    | 46×2    | 5 | 15 | 23,1    |
| spr_uniform_select2   | 36×2    | 2 | 15 | 18,1    |
| spr_uniform_select4   | 36×2    | 4 | —  | 7,10    |
| spr_watermark         | 960×540 | 1 | 15 | 480,270 |

## 17 Method and trust

The build was obtained as a public GameMaker HTML5 export and parsed, not played and eyeballed. The asset manifest inside `RetroBowl.js` carries, per sprite: name, authored width and height, origin, bounding box, playback speed, and the index of every frame into a 637-entry texture table that gives each frame's page, position, crop and untrimmed size. Sprite sizes were cross-checked against the atlas crops and against sprites whose names encode their own dimensions (`spr_box_128×128` is 128×128 in both places). Room and camera sizes come from the room definitions; the shader is quoted verbatim; the key colours are the literal uniform values set in the calling code; fonts are the typeface names recorded in the font definitions.

Two things in this document are not measurement. The **reading of intent** — why frames are spent where they are, why the crowd is a texture — is analysis. The **Retro Buhurt proposals** in §8 are recommendations, with the arithmetic behind each one shown so it can be checked rather than taken. The Retro Buhurt figures themselves are read out of the current source.

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